Morning Paper Report - Tuesday, July 28, 2026

Recent, non-repeated papers matched to visual working memory, visual search, modeling, working-memory representation, neural-network memory models, AI for human memory, and egocentric predictive-coding memory themes.

1. A Hierarchical Bayesian Analysis of Memory Load Effects on Visual Working Memory Precision

Erfan Jaripour
2026

Source: Crossref; matched terms: visual working memory, working memory, precision, computational model, computational models, modeling

Link: https://doi.org/10.21203/rs.3.rs-10469232/v1

Goal: Abstract Visual working memory is limited in both capacity and precision, and computational models increasingly aim to characterize how memory representations vary with increasing load and across individuals

Method: Abstract Visual working memory is limited in both capacity and precision, and computational models increasingly aim to characterize how memory representations vary with increasing load and across individuals

Results: Results showed that memory precision decreased systematically as memory load increased and that participants exhibited substantial differences in baseline precision

Conclusion: Robustness analyses demonstrated that the primary conclusions remained stable across alternative prior specifications, sampling configurations, and posterior initializations These findings highlight the value of hierarchical Bayesian approaches for modeling individual variability in visual working-memory performance and demonstrate the utility of reproducible computational workflows for evaluating statistical...

2. Explainable AI Models for Decoding Emotional Subtexts on Social Media

Dost Muhammad, Iftikhar Ahmed, Khwaja Naveed, Malika Bendechache
Complex, 2025

Source: Semantic Scholar; matched terms: precision, neural network, recurrent neural network, deep learning, artificial intelligence, ai model

Link: https://www.semanticscholar.org/paper/b757f0e6533a1e67d589c0137fbfb376c93442bd

Goal: Social media platforms, such as X (formerly Twitter), provide users with concise but impactful tools to express their views and feelings

Method: Numerous machine learning and deep learning approaches have been presented lately for optimal emotion detection and sentiment analysis of these tweets

Results: Subsequently, we integrate the explainable artificial intelligence (XAI) approaches, namely, local interpretable model■agnostic explanations (LIME) and SHapely Additive exPlanation (SHAP) to explain the decision■making process behind the proposed model’s prediction Applying these XAI techniques not only boosts the proposed model’s transparency but also reinforces its reliability in accurately processing and...

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3. From perception to action : Integrating cognitive systems with deep AI models for enhanced human-like intelligence

G. Shwetha, Maitri J Patel, Monica Bhutani, Nagamani Bhutani, H. S. Gururaja, Rijuta Joshi, et al.
Journal of Information & Optimization Sciences, 2026

Source: Semantic Scholar; matched terms: neural network, neural networks, recurrent neural network, transformer, deep learning, ai model

Link: https://www.semanticscholar.org/paper/1a1ff7d4679ca551879f45bce75a107c5f3556bc

Goal: The use of AI with cognitive technologies is dramatically reshaping the field of making intelligent machines

Method: Applying information from neuroscience and cognitive psychology, we design our model by using Vision Transformers (ViT) for vision tasks, Recurrent Neural Networks (RNN) to represent memory, attention mechanisms for studying focus and language models that generate output

Results: This work establishes the foundation for agents that can move through their surroundings as humans do and respond carefully to what is happening

Conclusion: The suggested framework is a step towards creating AI systems that function openly, are strong and process information similarly to human minds in real-world scenarios.

4. eLife Assessment: Neural dynamics of visual working memory representation during sensory distraction

Gui Xue
2025

Source: Crossref; matched terms: visual working memory, working memory representation, working memory

Link: https://doi.org/10.7554/elife.99290.3.sa2

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

Method: Method details were not available in the retrieved metadata.

Results: Result details were not available in the retrieved metadata.

Conclusion: Open the linked record to inspect the full abstract or paper before journal-club use.

5. Chunking in Visual Working Memory changes the Guidance of Attention in a Visual Search

Logan Doyle, Susanne Ferber
Journal of Vision, 2024

Source: Crossref; matched terms: visual working memory, visual search, working memory

Link: https://doi.org/10.1167/jov.24.10.678

Goal: The title and metadata suggest relevance to Lila's working-memory and attention topics, but no abstract was available from the source API.

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